

Module 4: ASG And ELB Assignment

Problem Statement: You work for XYZ Corporation that uses on premise solutions and a limited number of systems. With the increase in requests in their application, the load also increases. So, to handle the load the corporation has to buy more systems almost on a regular basis. Realizing the need to cut down the expenses on systems, they decided to move their infrastructure to AWS.

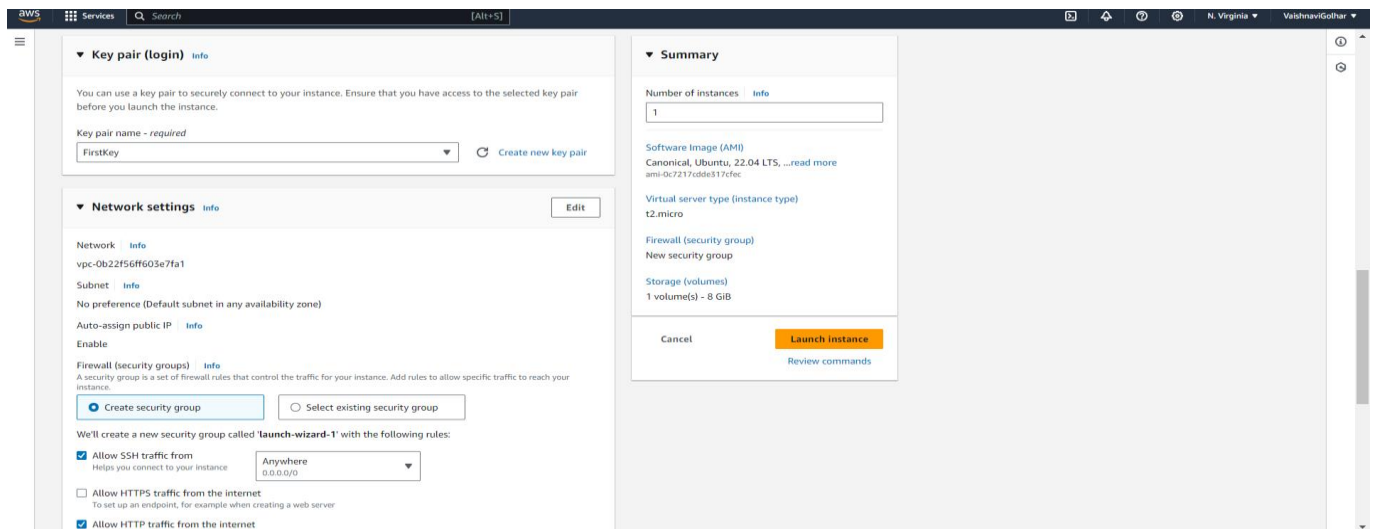
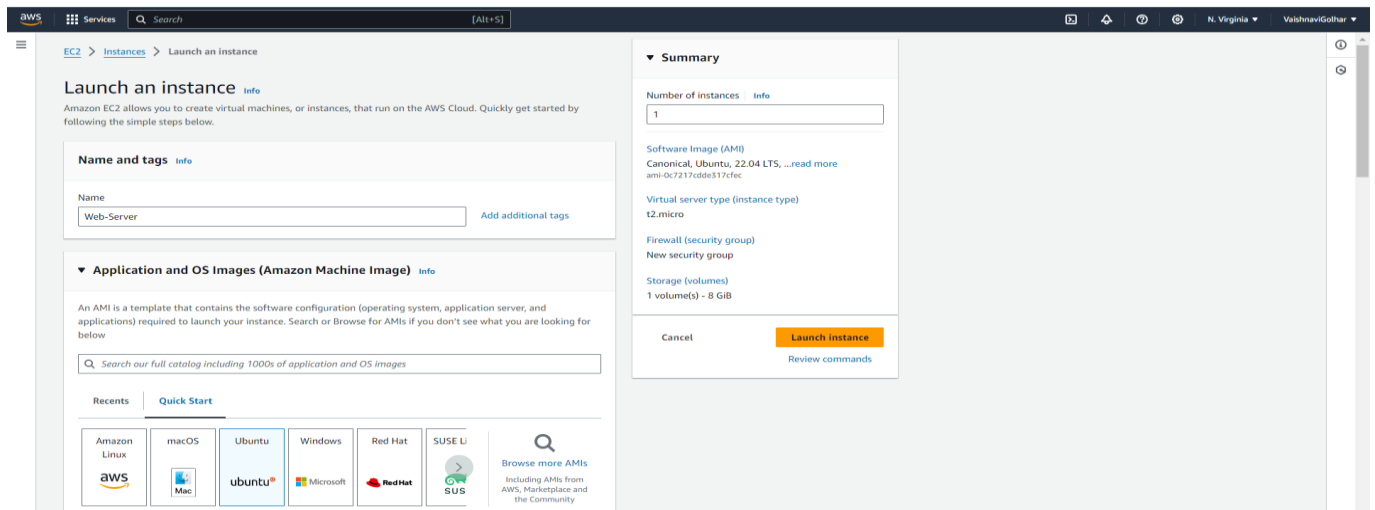
Tasks To Be Performed:

- 1. Manage the scaling requirements of the company by:**
 - a. Deploying multiple compute resources on the cloud as soon as the load increases and the CPU utilization exceeds 80%**
 - b. Removing the resources when the CPU utilization goes under 60%**

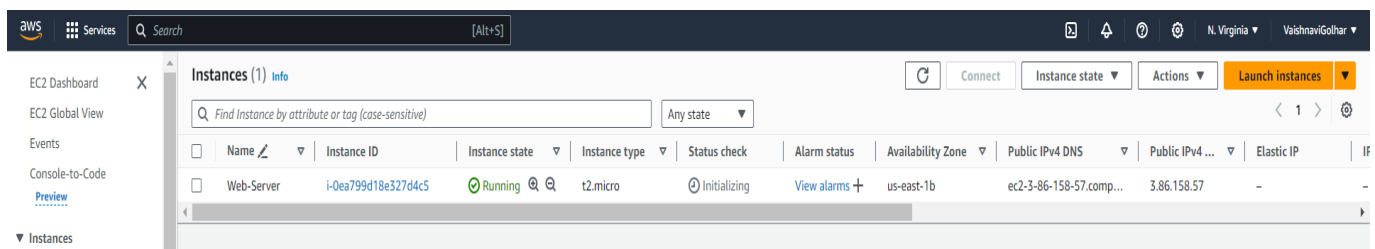
- 2. Create a load balancer to distribute the load between compute resources.**
- 3. Route the traffic to the company's domain.**

Solution:

- 1. Open up the AWS Management Console**
- 2. Check for the region [us-east-1(N. Virginia)]**
- 3. Search for EC2 in the search box**
- 4. Click on instances to go to the EC2 console**
- 5. Click on Launch Instance and setup the instance for Ubuntu OS:**
 - a. Name: Web-Server**
 - b. AMI: QuickStart >> Ubuntu OS [Any version which is free tier eligible]**
 - c. Instance type: t2. micro [free tier eligible]**
 - d. Key-pair: Create a key pair [rsa and. pem]**
 - e. Security group: Default**



6. Check the instance running condition.



7. Connect the instance.

8. Now connect Web-Sever. (Ubuntu)

a. Update sudo apt-get update.

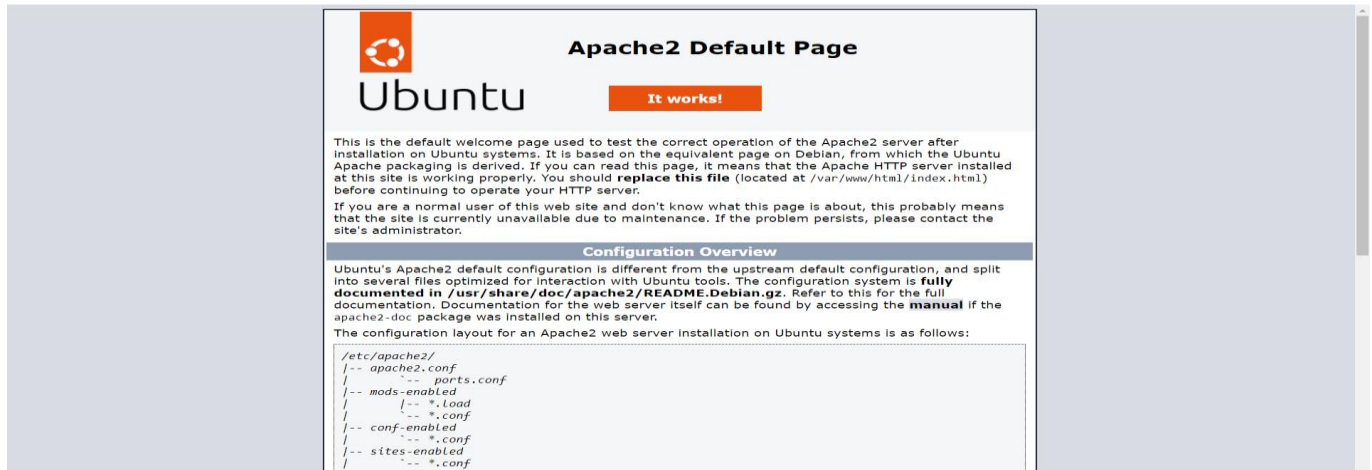
```
aws Services Search [Alt+S]
ubuntu@ip-172-31-85-31:~$ sudo apt-get update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates InRelease [119 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports InRelease [109 kB]
Get:4 http://security.ubuntu.com/ubuntu jammy-security InRelease [110 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [14.1 MB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/universe Translation-en [5652 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 c-n-f Metadata [286 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [217 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/multiverse Translation-en [112 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/multiverse amd64 c-n-f Metadata [8372 B]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [1412 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [277 kB]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [1490 kB]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/restricted Translation-en [249 kB]
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1049 kB]
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/universe Translation-en [237 kB]
Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 c-n-f Metadata [22.1 kB]
Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [42.1 kB]
Get:19 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/multiverse Translation-en [10.1 kB]
Get:20 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 c-n-f Metadata [472 B]
Get:21 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [41.7 kB]
Get:22 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/main Translation-en [10.5 kB]
Get:23 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/main amd64 c-n-f Metadata [388 B]
Get:24 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/restricted amd64 c-n-f Metadata [116 B]
Get:25 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [24.3 kB]
Get:26 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/universe Translation-en [16.5 kB]
Get:27 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 c-n-f Metadata [644 B]
Get:28 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/multiverse amd64 c-n-f Metadata [116 B]
Get:29 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [1192 kB]
Get:30 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [217 kB]
Get:31 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [1452 kB]
Get:32 http://security.ubuntu.com/ubuntu jammy-security/restricted Translation-en [239 kB]
Get:33 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [842 kB]
Get:34 http://security.ubuntu.com/ubuntu jammy-security/universe Translation-en [161 kB]
Get:35 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 c-n-f Metadata [16.8 kB]
Get:36 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [37.1 kB]
Get:37 http://security.ubuntu.com/ubuntu jammy-security/multiverse Translation-en [7476 B]

i-Oea799d18e327d4c5 (Web-Server)
Public IPs: 3.86.158.57 Private IPs: 172.31.85.31
```

b. Install Apache web: sudo apt-get install apache2 -y

```
aws Services Search [Alt+S]
ubuntu@ip-172-31-85-31:~$ sudo apt-get install apache2 -y
```

c. To check open new tab copy and paste the public IP to browser APACHE 2 WEB.

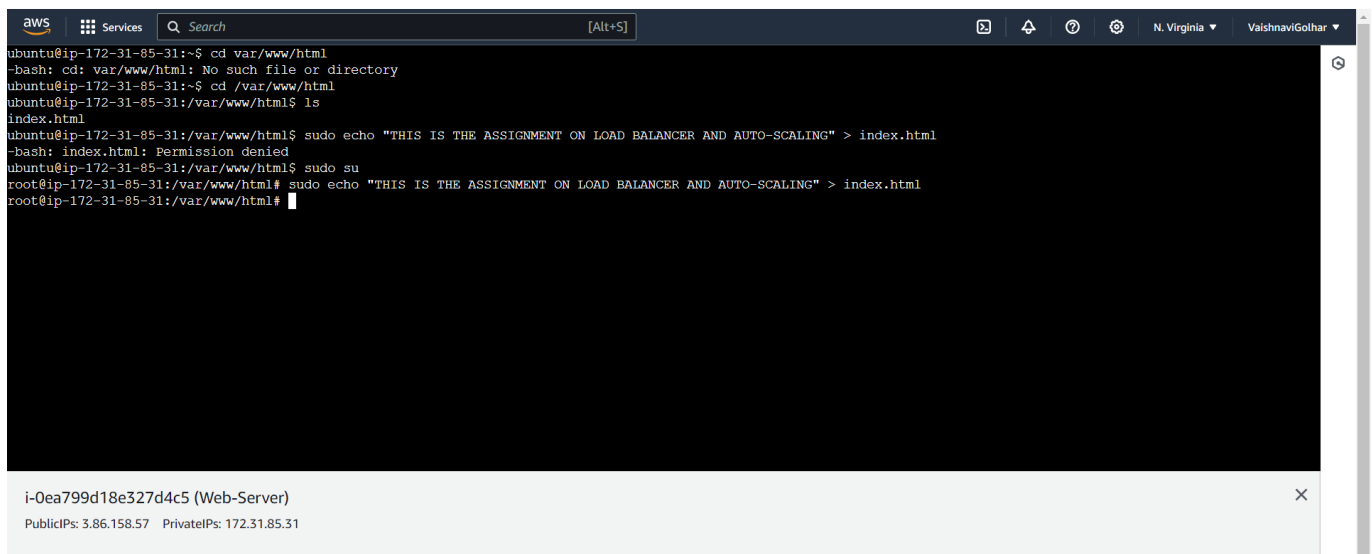


d. To consume it to our own name as

`cd /var/www/html`

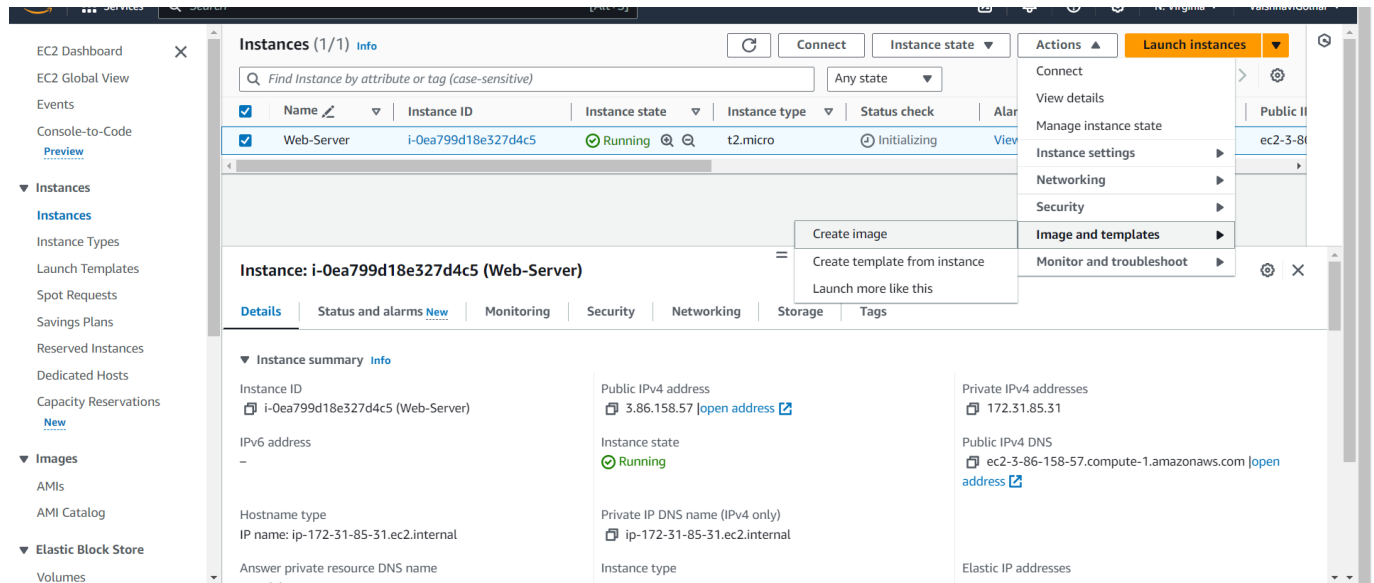
`sudo su`

`echo "THIS IS THE ASSIGNMENT ON LOAD
BALANCER AND AUTO SCALING" > index.html`



THIS IS THE ASSIGNMENT ON LOAD BALANCER AND AUTO-SCALING

9. Create an Image of the Web-Server instance.



The screenshot shows the AWS Management Console interface. On the left, the navigation pane includes 'EC2 Dashboard', 'EC2 Global View', 'Events', 'Console-to-Code', 'Instances', 'Instance Types', 'Launch Templates', 'Spot Requests', 'Savings Plans', 'Reserved Instances', 'Dedicated Hosts', 'Capacity Reservations', 'Images', 'AMI Catalog', and 'Elastic Block Store'. The main content area displays the 'Instances (1/1)' page. A table lists the instance 'Web-Server' with ID 'i-0ea799d18e327d4c5', state 'Running', and type 't2.micro'. The 'Actions' menu is open, showing options like 'Connect', 'View details', 'Manage instance state', 'Instance settings', 'Networking', 'Security', 'Image and templates', and 'Monitor and troubleshoot'. The 'Image and templates' option is highlighted, and a sub-menu is visible with 'Create image', 'Create template from instance', and 'Launch more like this'. Below the table, the 'Details' tab for the selected instance is shown, displaying various attributes such as Instance ID, Public IPv4 address, Private IPv4 addresses, Instance state, Hostname type, Private IP DNS name, and Elastic IP addresses.

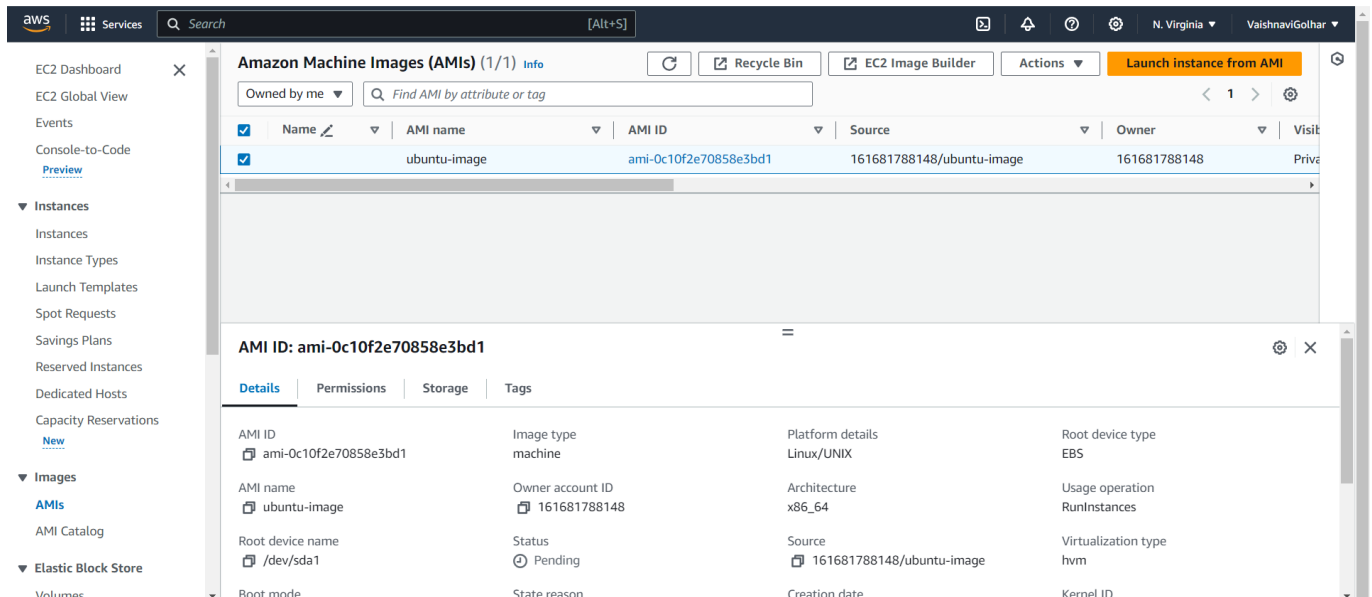
Name	Instance ID	Instance state	Instance type	Status check	Alarm
Web-Server	i-0ea799d18e327d4c5	Running	t2.micro	Initializing	View

Instance: i-0ea799d18e327d4c5 (Web-Server)

Instance summary

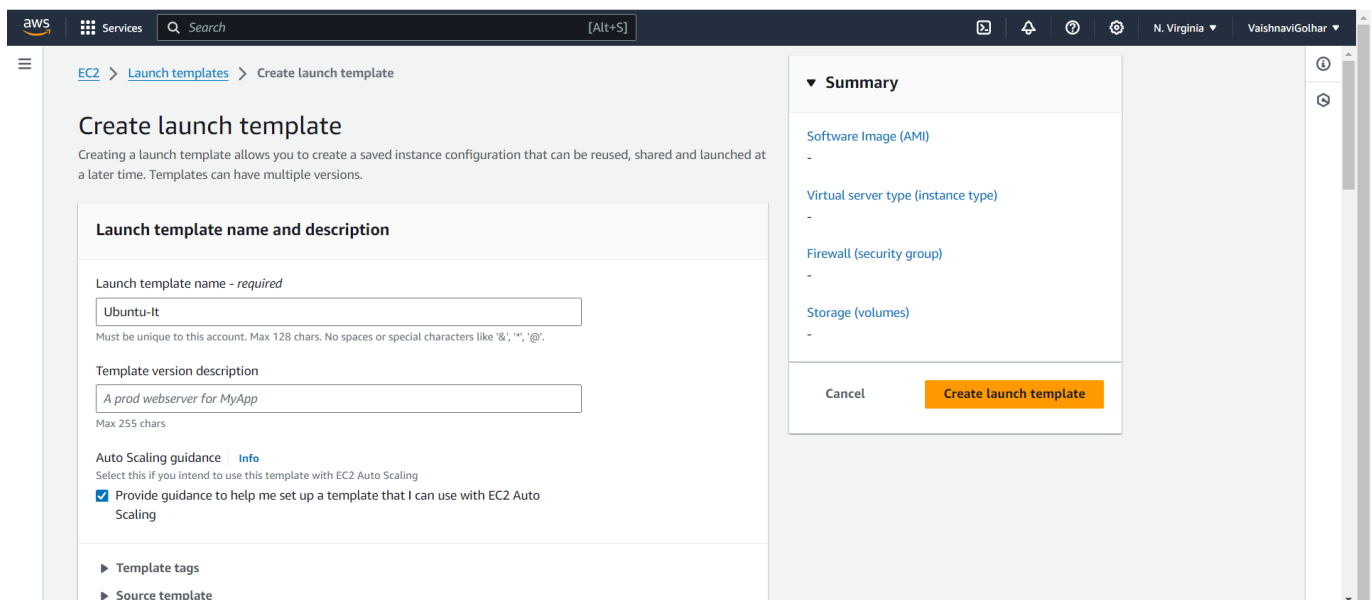
Instance ID	Public IPv4 address	Private IPv4 addresses
i-0ea799d18e327d4c5 (Web-Server)	3.86.158.57 open address	172.31.85.31
IPv6 address	Instance state	Public IPv4 DNS
-	Running	ec2-3-86-158-57.compute-1.amazonaws.com open address
Hostname type	Private IP DNS name (IPv4 only)	
IP name: ip-172-31-85-31.ec2.internal	ip-172-31-85-31.ec2.internal	
Answer private resource DNS name	Instance type	Elastic IP addresses
ip-172-31-85-31.ec2.internal	t2.micro	

10. Check in Image tab of AMIs with name of Ubuntu-Image

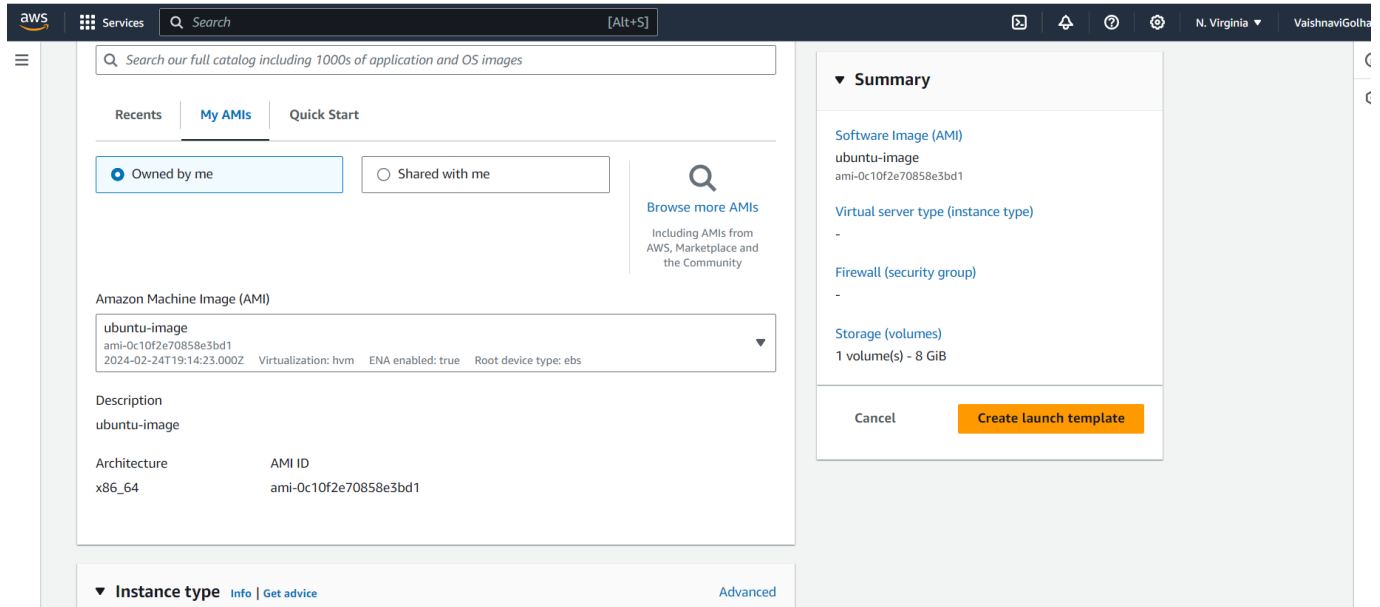


11. Go to the launch template.

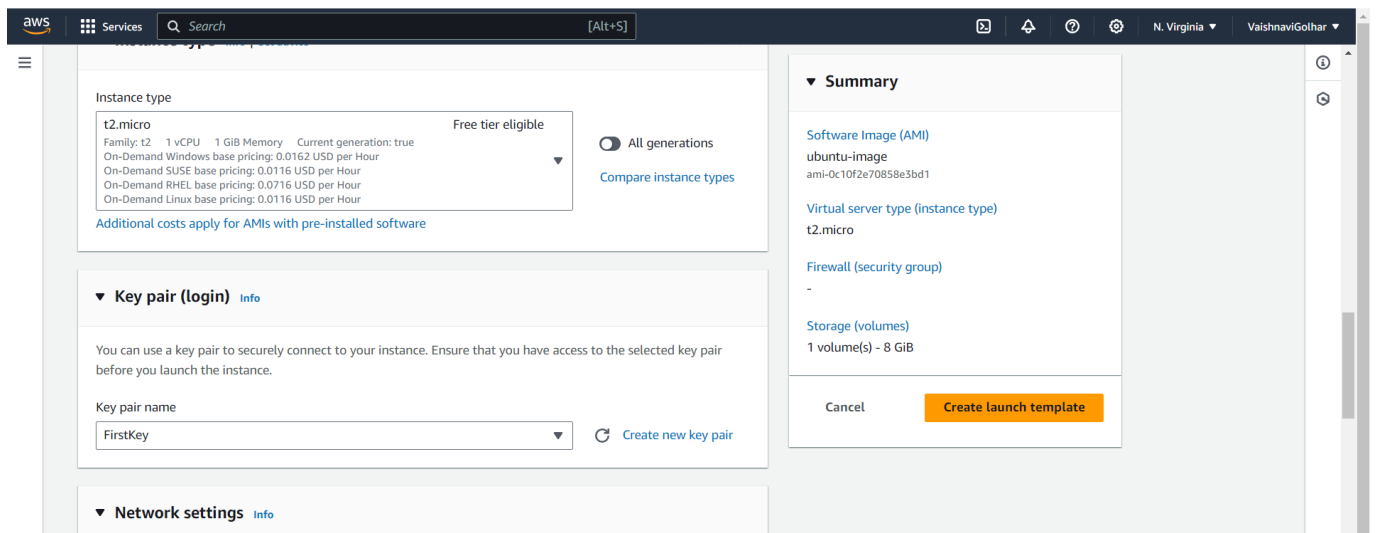
a. Launch template with a name ubuntu-It



b. Choose AMI's which is created by us in the AMIs (Owned by me)



c. With Instance type t2.micro and Key pair



d. With default security group, create launch template.

Network settings Info

Subnet Info
Don't include in launch template [Create new subnet](#)

When you specify a subnet, a network interface is automatically added to your template.

Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Select existing security group ☐ Create security group

Security groups Info
Select security groups

launch-wizard-1 sg-0e0f54b712a103ca2 X
VPC: vpc-0b22f56ff603e7fa1 [Compare security group rules](#)

► Advanced network configuration

Storage (volumes) Info

EBS Volumes [Hide details](#)

Summary

Software Image (AMI)
ubuntu-image
ami-0c10f2e70858e3bd1

Virtual server type (instance type)
t2.micro

Firewall (security group)
launch-wizard-1

Storage (volumes)
1 volume(s) - 8 GiB

Cancel [Create launch template](#)

e. Launch templates created.

Delete Launch Template Request Succeeded

Launch Templates (1/1) Info

[Search](#) [Create launch template](#)

Launch Template ID	Launch Template Name	Default Version	Latest Version	Create Time	Created ...
lt-0cf467c77fb4694e3	Ubuntu-It	1	1	2024-02-24T19:21:44.000Z	arn:aws:ia...

Ubuntu-It (lt-0cf467c77fb4694e3)

Launch template details [Actions](#) [Delete template](#)

Launch template ID	Launch template name	Default version	Owner
lt-0cf467c77fb4694e3	Ubuntu-It	1	arn:aws:iam::161681788148:root

Launch template version details [Actions](#) [Delete template version](#)

12. Create Auto-scaling group.

a. With a name ubuntu-asg and choose the launch template as ubuntu-lt.

The screenshot shows the AWS Management Console interface for creating an Auto Scaling group. The left sidebar contains a navigation menu with steps: Step 2 (Choose instance launch options), Step 3 - optional (Configure advanced options), Step 4 - optional (Configure group size and scaling), Step 5 - optional (Add notifications), Step 6 - optional (Add tags), and Step 7 (Review). The main content area is titled 'Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.' It features a 'Name' section with a text input field containing 'ubuntu-asg' and a 'Launch template' section with a dropdown menu set to 'Ubuntu-lt'. A blue information box states: 'For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.'

b. Network with a default VPC and for maximum availability choose 2-3 subnets click next

The screenshot shows the 'Network' step in the AWS Management Console. It includes a 'VPC' section with a dropdown menu set to 'vpc-0b22f56ff603e7fa1' and an 'Availability Zones and subnets' section with a dropdown menu set to 'Select Availability Zones and subnets'. Below these, three subnets are listed: 'us-east-1a | subnet-0d55ece629f34b37c', 'us-east-1b | subnet-0a7944fdc1d433250', and 'us-east-1c | subnet-091853ce987a0a3b6'. Each subnet entry has a close button (X) and a refresh button (circular arrow). A 'Create a subnet' link is visible at the bottom.

c. Configure advanced options as default click next.

The screenshot shows the 'Configure advanced options' step in the AWS Management Console. The left sidebar lists steps 1 through 7, with step 3 'Configure advanced options' highlighted. The main content area is titled 'Configure advanced options - optional' and includes an 'Info' icon. It contains two sections: 'Load balancing' and 'VPC Lattice integration options'. In the 'Load balancing' section, the 'No load balancer' option is selected. The 'VPC Lattice integration options' section is currently empty.

Configure advanced options - optional [Info](#)

Integrate your Auto Scaling group with other services to distribute network traffic across multiple servers using a load balancer or to establish service-to-service communications using VPC Lattice. You can also set options that give you more control over health check replacements and monitoring.

Load balancing [Info](#)

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

- ☒ **No load balancer**
Traffic to your Auto Scaling group will not be fronted by a load balancer.
- ☐ **Attach to an existing load balancer**
Choose from your existing load balancers.
- ☐ **Attach to a new load balancer**
Quickly create a basic load balancer to attach to your Auto Scaling group.

VPC Lattice integration options [Info](#)

To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and helps you connect and manage your applications across compute services in AWS.

Select VPC Lattice service to attach

d. Configure group size as desired capacity: 2, Max: 4 and Min: 1

The screenshot shows the 'Configure group size and scaling' step in the AWS Management Console. The left sidebar lists steps 1 through 7, with step 4 'Configure group size and scaling' highlighted. The main content area is titled 'Configure group size and scaling - optional' and includes an 'Info' icon. It contains two sections: 'Group size' and 'Scaling'. In the 'Group size' section, the 'Desired capacity' is set to 2. In the 'Scaling' section, the 'Min desired capacity' is set to 1 and the 'Max desired capacity' is set to 4. A right-hand sidebar titled 'Configure group size and scaling' provides additional information and a 'Learn more' link.

Configure group size and scaling - optional [Info](#)

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size [Info](#)

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances) ▾

Desired capacity

Specify your group size.

2

Scaling [Info](#)

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity: 1

Max desired capacity: 4

Configure group size and scaling

If you have an application that needs more capacity, you can increase the size of your Auto Scaling group now by updating the value for **Desired capacity** under **Group size**.

You can add a scaling policy now or wait until the Auto Scaling group is created and add one later. With scaling policies, you can scale your Amazon EC2 capacity automatically. When there is more load, scaling policies can launch more instances (referred to as *scaling out*). When load decreases, scaling policies can terminate unneeded instances (referred to as *scaling in*).

[Learn more](#)

[Scale the size of your Auto Scaling group](#)

e. Add notification -optional click next

The screenshot shows the 'Add notifications - optional' step in the AWS console. On the left, a sidebar lists steps 1 through 7, with 'Add notifications' highlighted as Step 5. The main area has the title 'Add notifications - optional' and a description: 'Send notifications to SNS topics whenever Amazon EC2 Auto Scaling launches or terminates the EC2 instances in your Auto Scaling group.' Below this is an 'Add notification' button. At the bottom are 'Cancel', 'Skip to review', 'Previous', and 'Next' buttons. On the right, a panel titled 'Add notifications' explains that notifications help keep users informed of changes by publishing to an Amazon SNS topic. It also includes a 'Learn more' link.

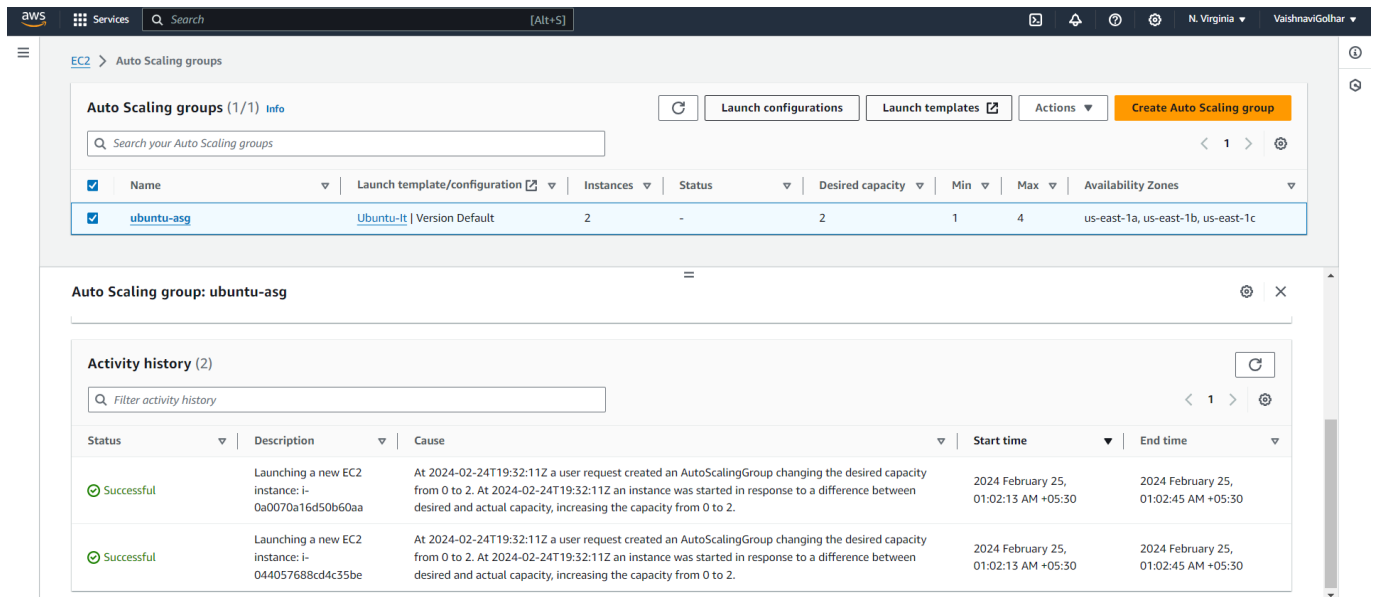
f. Add tags -optional click next

The screenshot shows the 'Add tags - optional' step in the AWS console. The sidebar on the left highlights 'Add tags' as Step 6. The main area has the title 'Add tags - optional' and a description: 'Add tags to help you search, filter, and track your Auto Scaling group across AWS. You can also choose to automatically add these tags to instances when they are launched.' Below the description is a blue information box stating: 'You can optionally choose to add tags to instances (and their attached EBS volumes) by specifying tags in your launch template. We recommend caution, however, because the tag values for instances from your launch template will be overridden if there are any duplicate keys specified for the Auto Scaling group.' Underneath is a section titled 'Tags (0)' with an 'Add tag' button and a note '50 remaining'. At the bottom are 'Cancel', 'Previous', and 'Next' buttons.

g. Review all the steps and create auto scaling group.

The screenshot shows the 'Review' step in the AWS console. The sidebar on the left highlights 'Review' as Step 7. The main area displays a summary of the configuration: 'Replacement strategy' (No policy), 'Rolling update' (No policy), 'Rolling update' (No policy), 'Instance scale-in protection' (Enabled), 'Step 5: Add notifications' (No notifications), and 'Step 6: Add tags' (No tags). Each section has an 'Edit' button. At the bottom are 'Cancel', 'Previous', and 'Create Auto Scaling group' buttons.

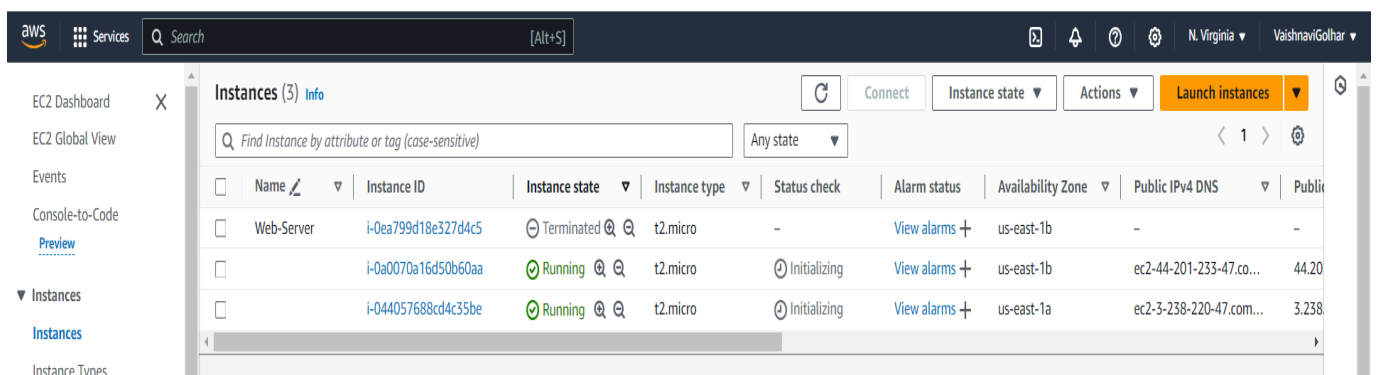
13. Auto Scaling Group is created successfully and check in the action tab desire= 2 instances are launched by auto scaling group.



The screenshot displays the AWS Management Console for the 'Auto Scaling groups' page. The 'ubuntu-asg' group is highlighted, showing a desired capacity of 2 instances. The activity history table below shows two successful launches of EC2 instances.

Status	Description	Cause	Start time	End time
Successful	Launching a new EC2 instance: i-0a0070a16d50b60aa	At 2024-02-24T19:32:11Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2024-02-24T19:32:11Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2024 February 25, 01:02:13 AM +05:30	2024 February 25, 01:02:45 AM +05:30
Successful	Launching a new EC2 instance: i-044057688cd4c35be	At 2024-02-24T19:32:11Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2024-02-24T19:32:11Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2024 February 25, 01:02:13 AM +05:30	2024 February 25, 01:02:45 AM +05:30

14. To verify this go to EC2 console and check it 2 desire instances are launched with initializing condition.



The screenshot displays the AWS Management Console for the 'Instances' page. Three instances are listed, with two in the 'Initializing' state.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IP
Web-Server	i-0ea799d18e327d4c5	Terminated	t2.micro	-	View alarms	us-east-1b	-	-
	i-0a0070a16d50b60aa	Running	t2.micro	Initializing	View alarms	us-east-1b	ec2-44-201-233-47.co...	44.20
	i-044057688cd4c35be	Running	t2.micro	Initializing	View alarms	us-east-1a	ec2-3-238-220-47.com...	3.238

15. Now open another browser and copy and paste the public IP. It displays the same message as that of AMIs image instance

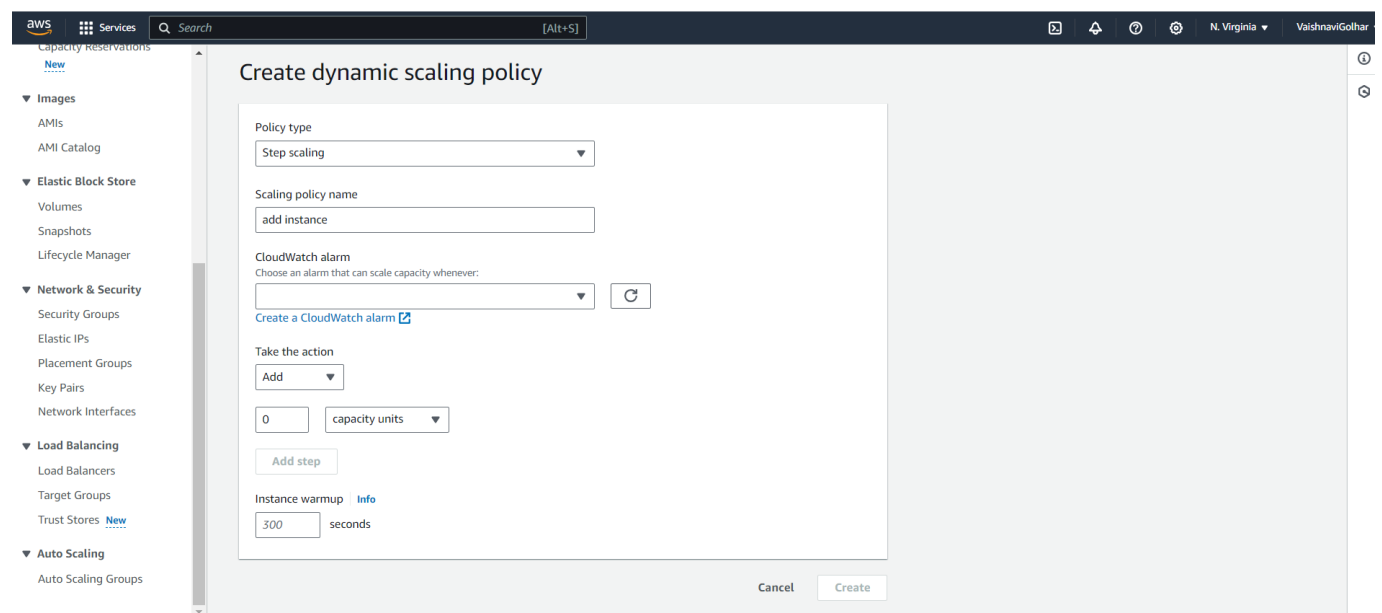


16. Create Dynamic scaling policy for adding and removing instances.

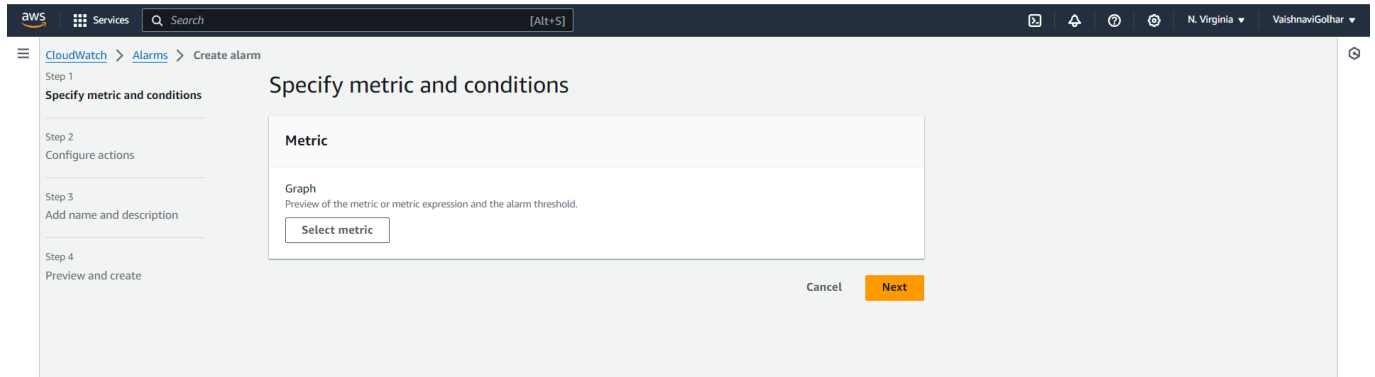
a. policy type as a step scaling

b. Scaling policy name: add instance and remove instance

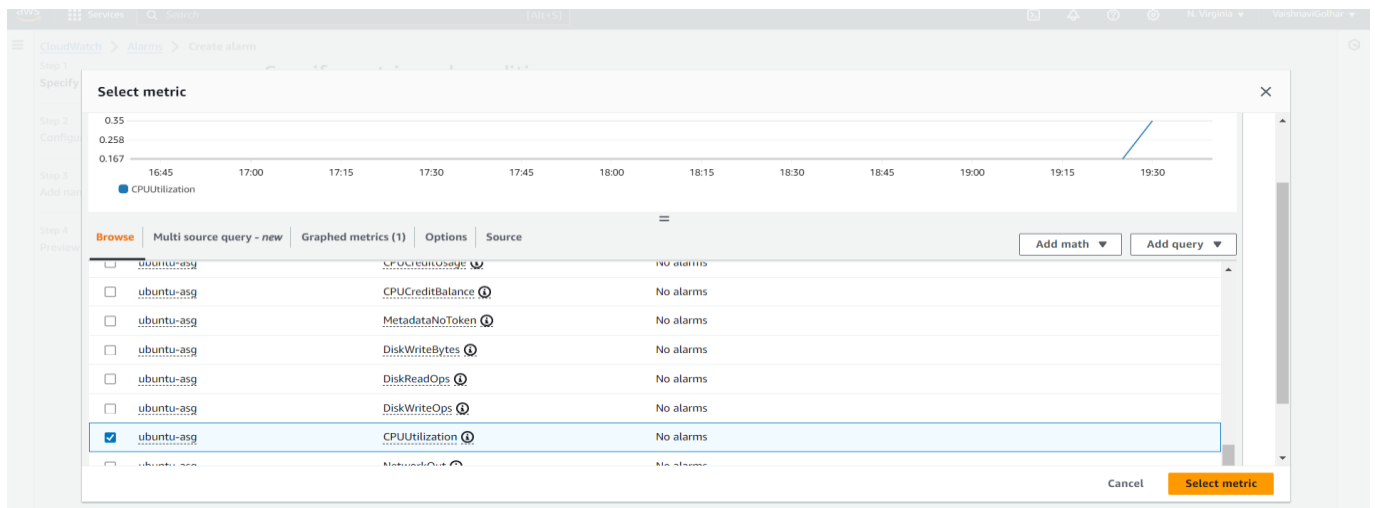
c. Using CloudWatch alarm



d. Create a CloudWatch alarm select metric click next.



e. Click on EC2 and Auto scaling then select CPU utilization of ubuntu-asg.



Specify metric and conditions

Metric Edit

Graph
This alarm will trigger when the blue line goes above the red line for 1 datapoints within 1 minute.

Percent

0.567

0.35

0.133

17:30 18:30 19:30

CPUUtilization

Namespace
AWS/EC2

Metric name
CPUUtilization

AutoScalingGroupName
ubuntu-asg

Statistic
Average

Period
1 minute

Conditions

Conditions

Threshold type

☒ Static
Use a value as a threshold

☐ Anomaly detection
Use a band as a threshold

Whenever CPUUtilization is...
Define the alarm condition.

☒ Greater
> threshold

☐ Greater/Equal
≥ threshold

☐ Lower/Equal
≤ threshold

☐ Lower
< threshold

than...
Define the threshold value.

80

Must be a number

► Additional configuration

Cancel Next

f. Similarly for <30% CPU Utilization with name remove ec2 instance.

Conditions

Threshold type

☒ Static
Use a value as a threshold

☐ Anomaly detection
Use a band as a threshold

Whenever CPUUtilization is...

Define the alarm condition.

☐ Greater
> threshold

☐ Greater/Equal
≥ threshold

☐ Lower/Equal
≤ threshold

☒ Lower
< threshold

than...

Define the threshold value.

30

Must be a number

► Additional configuration

Cancel **Next**

g. Check both alarm conditions.

CloudWatch

Alarms (2)

☐	Name	State	Last state update	Conditions	Actions
☐	RemoveEC2Instance	Insufficient data	2024-02-24 19:48:34	CPUUtilization < 30 for 1 datapoints within 1 minute	No actions
☐	AddEC2Instance	Insufficient data	2024-02-24 19:46:45	CPUUtilization > 80 for 1 datapoints within 1 minute	No actions

h. Creates for adding ec2

Policy type

Step scaling

Scaling policy name

add instance

CloudWatch alarm

Choose an alarm that can scale capacity whenever:

AddEC2Instance

Create a CloudWatch alarm [AddEC2Instance](#)

breaches the alarm threshold: CPUUtilization > 80 for 1 consecutive periods of 60 seconds for the metric dimensions:

AutoScalingGroupName = ubuntu-asg

Take the action

Add

1 capacity units when 80 ≤ CPUUtilization < +infinity

Add step

Instance warmup

60 seconds

Cancel **Create**

i. Creates for removing ec2 instance

Edit dynamic scaling policy

Policy type: Step scaling

Scaling policy name: RemoveEC2

CloudWatch alarm: RemoveEC2Instance
Choose an alarm that can scale capacity whenever:
breaches the alarm threshold: CPUUtilization < 30 for 1 consecutive periods of 60 seconds for the metric dimensions:
AutoScalingGroupName = ubuntu-asg

Take the action: Remove

1 capacity units when 30 >= CPUUtilization > -infinity

Buttons: Cancel, Update

j. Now two dynamic scaling policy are created for ubuntu-asg auto scaling group.

k. Remove ec2 instance is activity by dynamic scaling policy because CPU Utilization is less than 30%. So, one instance has to be removed.

Auto Scaling groups (1/1)

Name	Launch template/configuration	Instances	Status	Desired capacity	Min	Max	Avail...
ubuntu-asg	Ubuntu-It Version Default	3	-	2	1	4	us-eas...

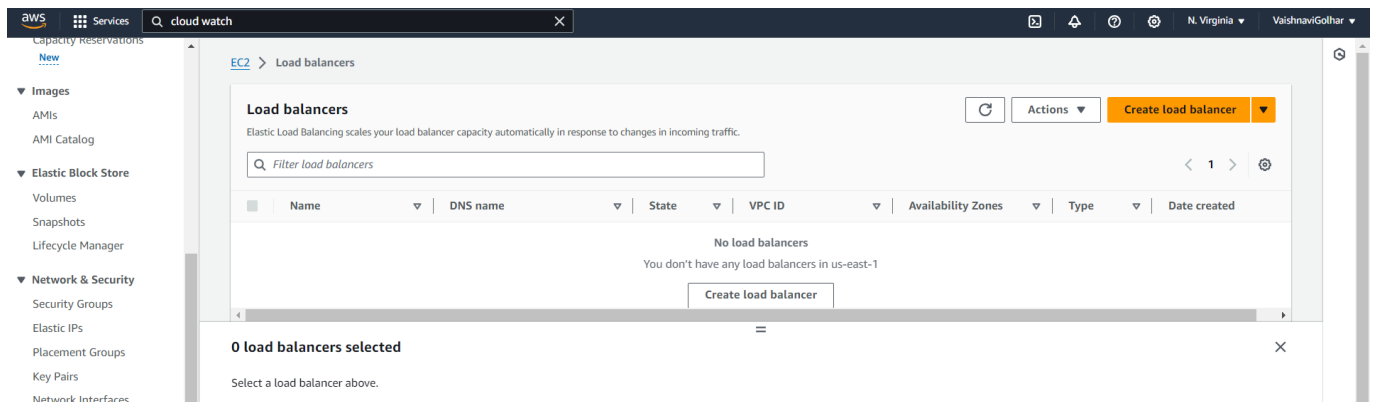
Auto Scaling group: ubuntu-asg

Status	Description	Cause	Start time	End time
In progress	Terminating EC2 instance: i-044057688cd4c35be	At 2024-02-24T19:58:56Z a monitor alarm RemoveEC2Instance in state ALARM triggered policy RemoveEC2 changing the desired capacity from 3 to 2. At 2024-02-24T19:59:03Z an instance was taken out of service in response to a difference between desired and actual capacity, shrinking the capacity from 3 to 2. At 2024-02-24T19:59:03Z instance i-044057688cd4c35be was selected for termination.	2024 February 25, 01:29:03 AM +05:30	
Successful	Launching a new EC2 instance: i-0a0070a16d50b60aa	At 2024-02-24T19:32:11Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2024-02-24T19:32:11Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2024 February 25, 01:02:13 AM +05:30	2024 February 25, 01:02:45 AM +05:30
Successful	Launching a new EC2 instance: i-044057688cd4c35be	At 2024-02-24T19:32:11Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2024-02-24T19:32:11Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2024 February 25, 01:02:13 AM +05:30	2024 February 25, 01:02:45 AM +05:30

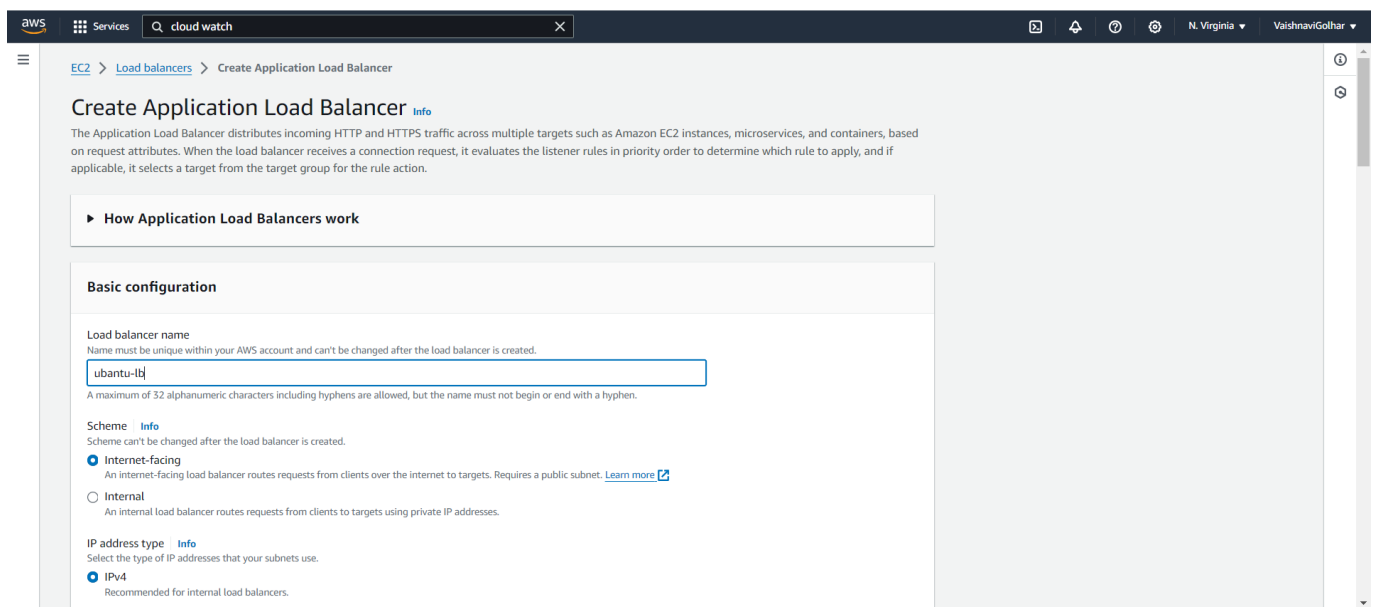
i. To verify this check in the EC2 console

17. Steps to create ELB

a. Create Load Balancer



b. Choose application load balancer with a name ubuntu-lb and with internet facing.



c. With IPv4 type of IP address and in network mapping select default VPC , for maximum availability chooses 3 subnets

The screenshot shows the 'Network mapping' configuration page in the AWS Management Console. The page is titled 'Network mapping' with an 'Info' link. Below the title, it states: 'The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.'

VPC [Info](#)
Select the virtual private cloud (VPC) for your targets or you can [create a new VPC](#). Only VPCs with an internet gateway are enabled for selection. The selected VPC can't be changed after the load balancer is created. To confirm the VPC for your targets, view your [target groups](#).

A dropdown menu shows the selected VPC: 'vpc-0b22f56ff603e7fa1' with an IPv4 address of '172.31.0.0/16'. A refresh button is next to it.

Mappings [Info](#)
Select at least two Availability Zones and one subnet per zone. The load balancer routes traffic to targets in these Availability Zones only. Availability Zones that are not supported by the load balancer or the VPC are not available for selection.

Three mappings are selected, each with a checkbox and a label:

- ☒ **us-east-1a (use1-az1)**
Subnet: 'subnet-0d55ece629f34b37c'
IPv4 address: 'Assigned by AWS'
- ☒ **us-east-1b (use1-az2)**
Subnet: 'subnet-0a7944fdc1d433250'
IPv4 address: 'Assigned by AWS'
- ☒ **us-east-1c (use1-az4)**
Subnet: 'subnet-091853ce987a0a3b6'

d. Create target group and Listener protocol HTTP and 80 port.

The screenshot shows the 'Listeners and routing' configuration page in the AWS Management Console. The page is titled 'Listeners and routing' with an 'Info' link. Below the title, it states: 'A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.'

▼ Listener HTTP:80 [Remove](#)

Protocol: HTTP, Port: 80, Default action: Forward to [Select a target group](#) [Create target group](#)

Listener tags - optional
Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

[Add listener tag](#)
You can add up to 50 more tags.

[Add listener](#)

► Load balancer tags - optional
Consider adding tags to your load balancer. Tags enable you to categorize your AWS resources so you can more easily manage them. The 'Key' is required, but 'Value' is optional. For example, you can have Key = production-webserver, or Key = webserver, and Value = production.

Optimize with service integrations - optional
Amazon can help you optimize your application by integrating AWS services with this load balancer at launch. You can also add these and other services after your load balancer is created to optimize the

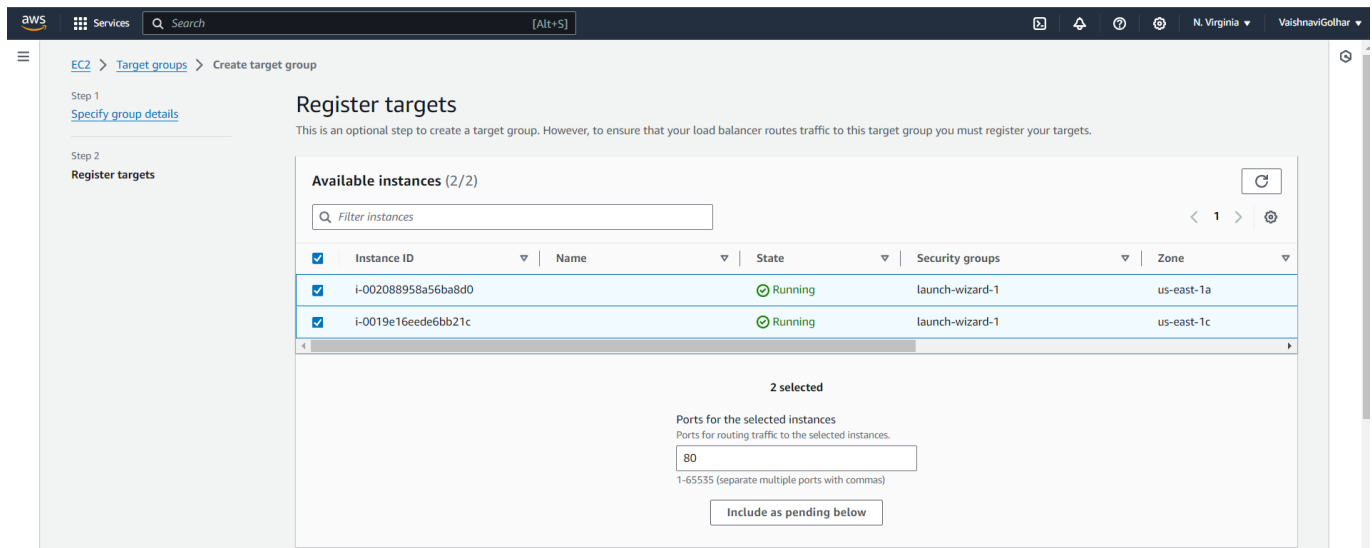
e. Choose the target type

The screenshot shows the 'Specify group details' page in the AWS Management Console. The breadcrumb navigation is 'EC2 > Target groups > Create target group'. The left sidebar shows 'Step 1: Specify group details' and 'Step 2: Register targets'. The main content area is titled 'Specify group details' with a subtitle 'Your load balancer routes requests to the targets in a target group and performs health checks on the targets.' Below this is a section 'Basic configuration' with a note 'Settings in this section can't be changed after the target group is created.' The 'Choose a target type' section has four radio button options: 'Instances' (selected), 'IP addresses', 'Lambda function', and 'Application Load Balancer'. The 'Instances' option is highlighted with a blue border and includes bullet points: 'Supports load balancing to instances within a specific VPC.' and 'Facilitates the use of [Amazon EC2 Auto Scaling](#) to manage and scale your EC2 capacity.'

f. With a target name as ubuntu-tg

The screenshot shows the 'Specify group details' page with the following fields filled out: 'Target group name' is 'ubuntu-tg' (with a note: 'A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.'). 'Protocol : Port' is set to 'HTTP' and '80' (with a range '1-65535'). 'IP address type' has 'IPv4' selected (with a note: 'Only targets with the indicated IP address type can be registered to this target group.' and details about the default network interface). 'IPv6' is unselected (with a note: 'Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)'). 'VPC' is set to 'vpc-0b22f56ff603e7fa1' (with a note: 'Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.'). 'Protocol version' has 'HTTP1' selected (with a note: 'Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.') and 'HTTP2' is unselected (with a note: 'Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.').

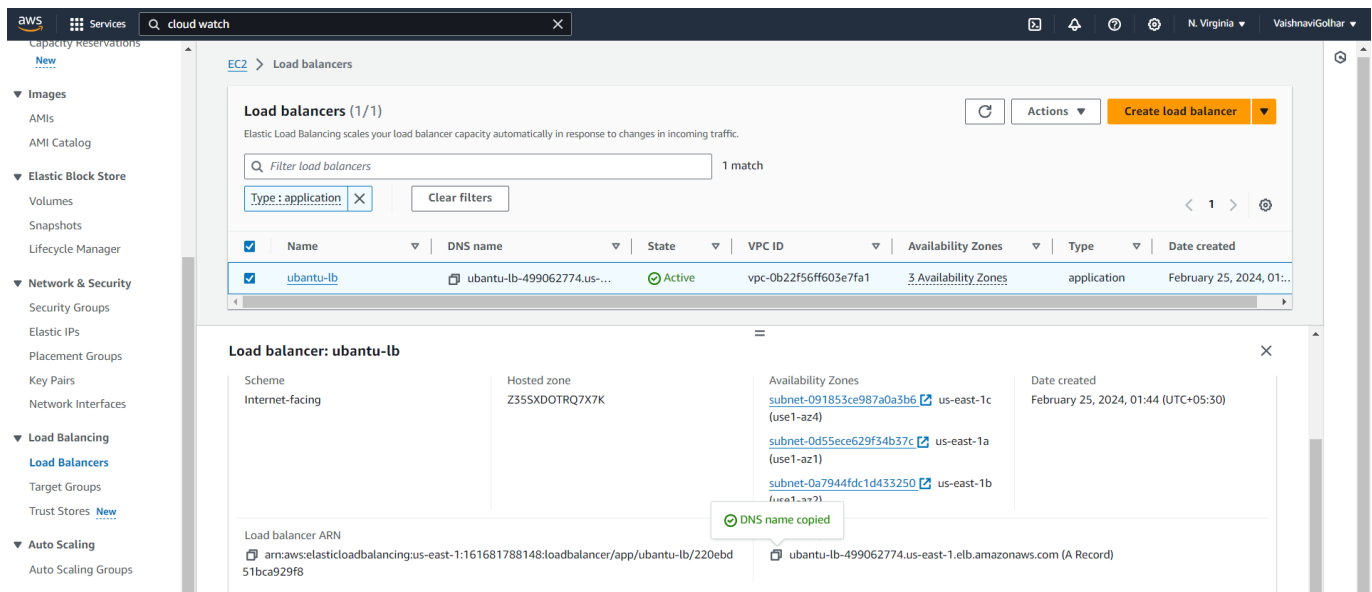
g. Select and register as instances



h. pending instance and create target group.

i. Select target group for load balancer

j. Create Load Balancer.



k. Check the healthy instance in the load balancer

l. Now copy the DNS of load balancer on the new browser and check apache3 web for two instances.



THIS IS THE ASSIGNMENT ON LOAD BALANCER AND AUTO-SCALING

